

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405483
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Kreikemeier; Sarah Margaret et al.

Piezo-based beam stabilization for cryogenic environments

Abstract

Aspects of the present disclosure relate generally to systems and methods for use in the implementation and/or operation of quantum information processing (QIP) systems, and more particularly, to various aspects of methods and systems for piezoelectric-based (or piezo-based) beam stabilization for cryogenic environments used in QIP systems.

Inventors: Kreikemeier; Sarah Margaret (San Diego, CA), Hudek; Kai Makoto (Hyattsville, MD)

Applicant: IonQ, Inc. (College Park, MD)

Family ID: 91028983

Assignee: IonQ, Inc. (College Park, MD)

Appl. No.: 18/164331

Filed: February 03, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240160031 A1	May. 16, 2024

Related U.S. Application Data

us-provisional-application US 63306689 20220204

Publication Classification

Int. Cl.: G02B27/64 (20060101); G02B26/08 (20060101)

U.S. Cl.:

Field of Classification Search

CPC: G21K (1/003); G02B (27/626); G03B (26/0883)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2019/0348251	12/2018	Monroe	N/A	H01J 37/08
2019/0363519	12/2018	Lochman	N/A	H01S 5/02326
2020/0028312	12/2019	Monroe	N/A	G02B 7/1827
2023/0108792	12/2022	Boege	N/A	H04N 23/55
2023/0200016	12/2022	Kreikemeier	62/49.1	F17C 3/085
2024/0086754	12/2023	Kreikemeier	N/A	G06N 10/40

Primary Examiner: Porta; David P

Attorney, Agent or Firm: ARENTFOX SCHIFF LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Patent Application No. 63/306,689, filed Feb. 4, 2022, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

(1) Aspects of the present disclosure relate generally to systems and methods for use in the implementation and/or operation of quantum information processing (QIP) systems.

BACKGROUND

(2) Trapped atoms are one of the leading implementations for quantum information processing or quantum computing. Atomic-based qubits may be used as quantum memories, as quantum gates in quantum computers and simulators, and may act as nodes for quantum communication networks. Qubits based on trapped atomic ions enjoy a rare combination of attributes. For example, qubits based on trapped atomic ions have very good coherence properties, may be prepared and measured with nearly 100% efficiency, and are readily entangled with each other by modulating their Coulomb interaction with suitable external control fields such as optical or microwave fields. These attributes make atomic-based qubits attractive for extended quantum operations such as quantum computations or quantum simulations.

(3) It is therefore important to develop new techniques that improve the design, fabrication, implementation, and/or control of different QIP systems used as quantum computers or quantum simulators, and particularly for those QIP systems that handle operations based on atomic-based qubits.

SUMMARY

(4) The following presents a simplified summary of one or more aspects to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects and is intended to neither identify key or critical elements of all aspects nor delineate the

scope of any or all aspects. Its sole purpose is to present some concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

(5) This disclosure describes various aspects of methods and systems for piezoelectric-based (or piezo-based) beam stabilization for cryogenic environments, including cryogenic environments used in QIP systems.

(6) In some aspects, a system for piezoelectric-based beam stabilization includes a laser beam source configured to produce a laser beam, a target, a lens assembly, and one or more piezoelectric transducers. The lens assembly includes a lens objective configured to produce an output beam from the laser beam and at least one optical element movably coupled to the lens assembly and configured to direct the laser beam to the target. The one or more piezoelectric transducers are coupled to at least one optical element and configured to dynamically reposition the at least one optical element in response to relative motion between the target and the lens assembly.

(7) In some aspects, a computer implemented method for performing piezoelectric-based beam stabilization for a cryogenic environment includes receiving, by a computing system including a processor and a memory, information indicative of a relative position of a target and a cryogenic lens assembly positioned inside the cryogenic environment; determining, by the computing system, an amount of displacement between the target and the lens assembly; generating, by the computing system, a repositioning command configured to actuate one or more piezoelectric elements coupled to a movable optical element of the lens assembly to reposition the movable optical element to direct an output beam produced by an objective lens of the lens assembly onto the target.

(8) To the accomplishment of the foregoing and related ends, the one or more aspects comprise the features hereinafter fully described and particularly pointed out in the claims. The following description and the annexed drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed, and this description is intended to include all such aspects and their equivalents.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosed aspects will hereinafter be described in conjunction with the appended drawings, provided to illustrate and not to limit the disclosed aspects, wherein like designations denote like elements, and in which:

(2) FIG. 1 illustrates a view of atomic ions a linear crystal or chain in accordance with aspects of this disclosure.

(3) FIG. 2 illustrates an example of a quantum information processing (QIP) system in accordance with aspects of this disclosure.

(4) FIG. 3 illustrates an example of a computer device in accordance with aspects of this disclosure.

(5) FIG. 4 illustrates a cryogenic lens assembly and a target positioned within a cryostat in accordance with aspects of this disclosure.

(6) FIG. 5 illustrates an example configuration in which piezoelectric actuators are configured to actuate a dove prism movably coupled to the cryogenic lens assembly to maintain alignment with the target in accordance with aspects of this disclosure.

(7) FIG. 6 illustrates an example of piezoelectric adjustments to the dove prism to correct misalignments and dynamically stabilize an output beam to a target in accordance with aspects of this disclosure.

(8) FIG. 7 illustrates another example of piezoelectric adjustments to the dove prism to correct misalignments and dynamically stabilize an output beam to a target in accordance with aspects of this disclosure.

(9) FIG. 8 illustrates an example of beam stabilization by showing the lens assembly as a fixed reference point and the target as shifting in accordance with aspects of this disclosure.

(10) FIG. 9 illustrates an example method of correcting misalignments and dynamically stabilizing an output beam to a target in accordance with aspects of this disclosure.

DETAILED DESCRIPTION

(11) The detailed description set forth below in connection with the appended drawings or figures is intended as a description of various configurations or implementations and is not intended to represent the only configurations or implementations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details or with variations of these specific details. In some instances, well known components are shown in block diagram form, while some blocks may be representative of one or more well known components.

(12) Cryogenic chambers may be used in various applications, including in some QIP systems such as QIP systems based on trapped-ion technologies or other atomic-based technologies. It is advantageous for high performance laser systems that interface with cryogenic chambers to be able to compensate for vibrations that occur within a cryostat. It is further advantageous for such systems to be able to be adjustable and/or repositionable by sufficient degrees of freedom to maintain alignment between optical components and/or targets inside the cryostat as the cryostat is cooled down to operating temperatures. For example, assemblies within the cryostat will undergo thermal contraction and will be exposed to vibrations produced by the pumping of gaseous or liquid helium necessary for maintaining the cryogenic environment. In many applications, optical components that are installed within a cryostat require precise alignment and stability relative to other internal and/or external components.

(13) Solutions to the issues described above are explained in more detail in connection with FIGS. 1-7, with FIGS. 1-3 providing a background of QIP systems or quantum computers, and more specifically, of atomic-based QIP systems or quantum computers such as trapped-ion QIP systems for example. Aspects of these solutions involve, for example, stabilizing the output beam of a cryogenic lens objective to a target using piezoelectric actuators and non-contact sensors for feedback. Typical applications often take an alternate approach of adjusting the position of the target or the position of the lens to maintain interferometric stability and alignment.

(14) FIG. 1 illustrates a diagram 100 with multiple atomic ions 106 (e.g., atomic ions 106a, 106b, . . . , 106c, and 106d) trapped in a linear crystal or chain 110 using a trap (the trap can be inside a vacuum chamber as shown in FIG. 2). The trap maybe referred to as an ion trap. The ion trap shown may be built or fabricated on a semiconductor substrate, a dielectric substrate, or a glass die or wafer (also referred to as a glass substrate). The atomic ions 106 may be provided to the trap as atomic species for ionization and confinement into the chain 110.

(15) In the example shown in FIG. 1, the trap includes electrodes for trapping or confining multiple atomic ions into the chain 110 that are laser-cooled to be nearly at rest. The number of atomic ions (N) trapped can be configurable and more or fewer atomic ions may be trapped. The atomic ions can be Ytterbium ions (e.g., ¹⁷¹Yb.^{sup.}+ ions), for example. The atomic ions are illuminated with laser (optical) radiation tuned to a resonance in ¹⁷¹Yb.^{sup.}+ and the fluorescence of the atomic ions is imaged onto a camera or some other type of detection device. In this example, atomic ions may be separated by about 5 microns (μm) from each other, although the separation may be smaller or larger than 5 μm. The separation of the atomic ions is determined by a balance between the external confinement force and Coulomb repulsion and does not need to be uniform. Moreover, in addition to atomic Ytterbium ions, neutral atoms, Rydberg atoms, different atomic ions or different species of atomic ions may also be used. The trap may be a linear RF Paul trap, but other types of confinement may also be used, including optical confinements. Thus, a confinement device may be based on different techniques and may hold ions, neutral atoms, or

Rydberg atoms, for example, with an ion trap being one example of such a confinement device. The ion trap may be a surface trap, for example.

(16) In some instances, to improve operational performance, the chain **110** may be placed inside a cryogenic environment such as the ones described herein.

(17) FIG. 2 shows a block diagram that illustrates an example of a QIP system **200** in accordance with various aspects of this disclosure. The QIP system **200** may also be referred to as a quantum computing system, a quantum computer, a computer device, a trapped ion system, or the like. The QIP system **200** may be part of a hybrid computing system in which the QIP system **200** is used to perform quantum computations and operations and the hybrid computing system also includes a classical computer to perform classical computations and operations.

(18) Shown in FIG. 2 is a general controller **205** configured to perform various control operations of the QIP system **200**. Instructions for the control operations may be stored in memory (not shown) in the general controller **205** and may be updated over time through a communications interface (not shown). Although the general controller **205** is shown separate from the QIP system **200**, the general controller **205** may be integrated with or be part of the QIP system **200**. The general controller **205** may include an automation and calibration controller **280** configured to perform various calibration, testing, and automation operations associated with the QIP system **200**.

(19) The QIP system **200** may include an algorithms component **210** that may operate with other parts of the QIP system **200** to perform quantum algorithms or quantum operations, including a stack or sequence of combinations of single qubit operations and/or multi-qubit operations (e.g., two-qubit operations) as well as extended quantum computations. As such, the algorithms component **210** may provide instructions to various components of the QIP system **200** (e.g., to the optical and trap controller **220**) to enable the implementation of the quantum algorithms or quantum operations. The algorithms component **210** may receive information resulting from the implementation of the quantum algorithms or quantum operations and may process the information and/or transfer the information to another component of the QIP system **200** or to another device for further processing.

(20) The QIP system **200** may include an optical and trap controller **220** that controls various aspects of a trap **270** in a chamber **250**, including the generation of signals to control the trap **270**, and controls the operation of lasers and optical systems that provide optical beams that interact with the atoms or ions in the trap. When used to confine or trap ions, the trap **270** may be referred to as an ion trap. The trap **270**, however, may also be used to trap neutral atoms, Rydberg atoms, different atomic ions or different species of atomic ions. The lasers and optical systems can be at least partially located in the optical and trap controller **220** and/or in the chamber **250**. For example, optical systems within the chamber **250** may refer to optical components or optical assemblies.

(21) The QIP system **200** may include an imaging system **230**. The imaging system **230** may include a high-resolution imager (e.g., CCD camera) or other type of detection device (e.g., photomultiplier tube or PMT) for monitoring the atomic ions while they are being provided to the trap **270** and/or after they have been provided to the trap **270**. In an aspect, the imaging system **230** can be implemented separate from the optical and trap controller **220**, however, the use of fluorescence to detect, identify, and label atomic ions using image processing algorithms may need to be coordinated with the optical and trap controller **220**.

(22) In addition to the components described above, the QIP system **200** can include a source **260** that provides atomic species (e.g., a plume or flux of neutral atoms) to the chamber **250** having the trap **270**. When atomic ions are the basis of the quantum operations, that trap **270** confines the atomic species once ionized (e.g., photoionized). The trap **270** may be part of a processor or processing portion of the QIP system **200**. That is, the trap **270** may be considered at the core of the processing operations of the QIP system **200** since it holds the atomic-based qubits that are used to perform the quantum operations or simulations. At least a portion of the source **260** may be implemented separate from the chamber **250**.

(23) It is to be understood that the various components of the QIP system **200** described in FIG. 2 are described at a high-level for ease of understanding. Such components may include one or more sub-components, the details of which may be provided below as needed to better understand certain aspects of this disclosure.

(24) For example, at least parts of the chamber **250** may operate under cryogenic conditions. In one example, the chamber **250** may be a cryostat or comprise a cryostat to provide specific cryogenic conditions. The chamber **250** may support one or more stages. In one example, the chamber **250** may support a stage at 4 Kelvin. In another example, the chamber **250** may support a first stage or first environment at 40 Kelvin, and a second stage or second environment within the first stage at 4 Kelvin.

(25) Aspects of this disclosure may be implemented at least partially using the chamber **250** and/or components therein.

(26) Referring now to FIG. 3, illustrated is an example of a computer system or device **300** in accordance with aspects of the disclosure. The computer device **300** can represent a single computing device, multiple computing devices, or a distributed computing system, for example. The computer device **300** may be configured as a quantum computer (e.g., a QIP system), a classical computer, or to perform a combination of quantum and classical computing functions, sometimes referred to as hybrid functions or operations. For example, the computer device **300** may be used to process information using quantum algorithms, classical computer data processing operations, or a combination of both. In some instances, results from one set of operations (e.g., quantum algorithms) are shared with another set of operations (e.g., classical computer data processing). A generic example of the computer device **300** implemented as a QIP system capable of performing quantum computations and simulations is, for example, the QIP system **200** shown in FIG. 2.

(27) The computer device **300** may include a processor **310** for carrying out processing functions associated with one or more of the features described herein. The processor **310** may include a single or multiple set of processors or multi-core processors. Moreover, the processor **310** may be implemented as an integrated processing system and/or a distributed processing system. The processor **310** may include one or more central processing units (CPUs) **310a**, one or more graphics processing units (GPUs) **310b**, one or more quantum processing units (QPUs) **310c**, one or more intelligence processing units (IPUs) **310d** (e.g., artificial intelligence or AI processors), or a combination of some or all those types of processors. In one aspect, the processor **310** may refer to a general processor of the computer device **300**, which may also include additional processors **310** to perform more specific functions (e.g., including functions to control the operation of the computer device **300**).

(28) The computer device **300** may include a memory **320** for storing instructions executable by the processor **310** to carry out operations. The memory **320** may also store data for processing by the processor **310** and/or data resulting from processing by the processor **310**. In an implementation, for example, the memory **320** may correspond to a computer-readable storage medium that stores code or instructions to perform one or more functions or operations. Just like the processor **310**, the memory **320** may refer to a general memory of the computer device **300**, which may also include additional memories **320** to store instructions and/or data for more specific functions.

(29) It is to be understood that the processor **310** and the memory **320** may be used in connection with different operations including but not limited to computations, calculations, simulations, controls, calibrations, system management, and other operations of the computer device **300**, including any methods or processes described herein.

(30) Further, the computer device **300** may include a communications component **330** that provides for establishing and maintaining communications with one or more parties utilizing hardware, software, and services. The communications component **330** may also be used to carry communications between components on the computer device **300**, as well as between the

computer device **300** and external devices, such as devices located across a communications network and/or devices serially or locally connected to computer device **300**. For example, the communications component **330** may include one or more buses, and may further include transmit chain components and receive chain components associated with a transmitter and receiver, respectively, operable for interfacing with external devices. The communications component **330** may be used to receive updated information for the operation or functionality of the computer device **300**.

(31) Additionally, the computer device **300** may include a data store **340**, which can be any suitable combination of hardware and/or software, which provides for mass storage of information, databases, and programs employed in connection with the operation of the computer device **300** and/or any methods or processes described herein. For example, the data store **340** may be a data repository for operating system **360** (e.g., classical OS, or quantum OS, or both). In one implementation, the data store **340** may include the memory **320**. In an implementation, the processor **310** may execute the operating system **360** and/or applications or programs, and the memory **320** or the data store **340** may store them.

(32) The computer device **300** may also include a user interface component **350** configured to receive inputs from a user of the computer device **300** and further configured to generate outputs for presentation to the user or to provide to a different system (directly or indirectly). The user interface component **350** may include one or more input devices, including but not limited to a keyboard, a number pad, a mouse, a touch-sensitive display, a digitizer, a navigation key, a function key, a microphone, a voice recognition component, any other mechanism capable of receiving an input from a user, or any combination thereof. Further, the user interface component **350** may include one or more output devices, including but not limited to a display, a speaker, a haptic feedback mechanism, a printer, any other mechanism capable of presenting an output to a user, or any combination thereof. In an implementation, the user interface component **350** may transmit and/or receive messages corresponding to the operation of the operating system **360**. When the computer device **300** is implemented as part of a cloud-based infrastructure solution, the user interface component **350** may be used to allow a user of the cloud-based infrastructure solution to remotely interact with the computer device **300**.

(33) In connection with the systems described in FIGS. 1-3, this disclosure describes examples of beam stabilization in cryogenic environments including those used for QIP systems. In an example aspect, this disclosure describes mechanisms and configurations to maintain interferometric alignment between the output of a cryogenic lens and a target without adjusting the position of the cryogenic lens or the target. As used herein, the phrase “interferometric alignment” means that two components (e.g., the lens and the target) are aligned to within an optical wavelength of a laser beam used in an interferometer sensor. In some aspects, the aligned components are aligned to within less than a micron. In some aspects, the aligned components are aligned to within 10 nm. In some aspects, the aligned components are aligned to within 1 nm. As used herein, the phrase “cryogenic lens” refers to a lens that is positioned in the cryostat chamber **250**. Due to the mass of the cryogenic lens and the need to maintain dynamic stability of the cryogenic lens, the cryogenic lens is rigidly mounted to a mechanical structure within the cryostat. The target is also rigidly mounted to a separate mechanical structure within the cryostat. In some aspects, a number of complex interfaces may be used to maintain the target in a fixed position relative to the cryostat. While both the target and the cryogenic lens assembly are fixed within the cryostat, their positions relative to one another are not. For example, differences in thermal contraction and unique vibration responses of the target and lens contribute to interferometric instability and misalignment during operation of the cryostat. Thus, the mechanisms described herein enable components within a cryogenic environment (e.g., within a cryogenic chamber or cryostat) to be precisely aligned and stable relative to other internal or external components.

(34) FIG. 4 illustrates a cross-sectional view of a cryostat **400** that includes a housing **404** that

defines the cryostat chamber **250**, as described above with respect to FIG. 2, according to an example aspect. Moreover, the housing **404** also provides a vacuum that insulates the cryostat chamber **250** from the ambient temperature conditions surrounding the cryostat **400**. The cryostat housing **404** may include one or more viewports **408** that are configured to allow optical access to the cryostat chamber **250**.

(35) A helium flow line is configured to provide a flow of cooled helium to a cryogenic base plate to maintain the cryostat chamber **250** at cryostatic temperatures. In operation, the pumping of liquid or gaseous helium necessary for creating cryogenic conditions may introduce vibration into the cryogenic chambers **250**. In an example aspect, a first stage or intermediate cooling stage and a second stage or cryostatic stage are coupled to the cryogenic base plate. Moreover, a temperature of the first stage can be maintained at approximately 40 K and a temperature of the second stage can be maintained at approximately 4 K.

(36) In the illustrated aspect, a cryogenic lens assembly **412** and a target **416** are positioned within the cryostat chamber **250** and may be mounted to the cryostat stages to maintain the cryogenic lens assembly **412** and the target **416** at the temperature of the particular cryostat stage. In some aspects, the cryogenic lens assembly **412** and the target **416** may be coupled to different stages. For example, the cryogenic lens assembly **412** may be coupled to the first stage and the target **416** may be coupled to the second stage. In another aspect, the cryogenic lens assembly **412** and the target **416** may be coupled to the same stage.

(37) In this aspect, the cryogenic lens assembly **412** is configured to direct a laser beam **420** onto a particular portion of the target **416**. In the illustrated aspect, the laser beam **420** is produced by a source **424** positioned outside of the cryostat **400**. The source **424** is oriented such that the laser beam **420** can enter the chamber **250** through the viewport **408**. In some aspects, adjustment knobs positioned upstream of the cryogenic lens assembly **412** may be provided to tune (e.g., by an operator) the assembly **412** to ensure the beam **420** is aligned to an optical axis of a lens of the cryogenic lens assembly **412** before the beam **420** enters the cryostat chamber **250**. For example, the incoming beam **420** is aligned in pitch and yaw to the optical axis A (FIG. 5) of the cryogenic lens assembly **412**.

(38) The cryogenic lens assembly **412** may be mounted to a cryogenic lens assembly mounting structure **428**. The cryogenic lens assembly mounting structure **428** may be mounted to a cryogenic lens platform **432**, which in turn may be or include a portion of first or second stage of the cryostat **400**.

(39) The target **416** may include the ion trap **270**. The target **416** may be mounted to a target mounting structure **436**. In aspects in which the target **416** includes the ion trap **270**, the target mounting structure **436** may be a cold finger assembly to which the ion trap **270** is mounted. In other aspects, the target **416** may be or include detectors and/or optical components. In yet other aspects, the target **416** may be or include a particular qubit. The target mounting structure **436** may be mounted to a cryogenic target platform **440**, which in turn may be or include a portion of the first or second stage of the cryostat **400**. A distance between the cryogenic lens assembly **412** and the target **416** is indicated by the arrow **l**. As described in detail below, it is advantageous to maintain the distance **l** between the cryogenic lens assembly **412** and the target **416** to be substantially constant during operation of the cryostat **400**. In configurations in which the cryogenic lens assembly **412** and the target **416** are coupled to different stages of the cryostat **400**, the distance **l** between the cryogenic lens assembly **412** and the target **416** may be maintained substantially constant during operation of the cryostat **400** to compensate for relative vibrations between the different stages, as described in greater detail below.

(40) During operation of the cryostat **400**, the cryogenic lens platform **432** and the cryogenic target platform **440** typically each experience their own vibration spectrum in response to vibrations of the cryostat **400**. For example, schematic representations of the vibration spectrum of the cryogenic lens platform **432** is shown as **V1** and the vibration spectrum of the cryogenic target platform **440** is

shown as V2 in FIG. 4. The different vibration spectra of the cryogenic lens platform 432 and the cryogenic target platform 440 may cause relative motion between the cryogenic lens assembly 412 and the target 416. As a result, this relative motion may cause misalignment between the cryogenic lens assembly 412 and the target 416. In configurations in which the cryogenic lens platform 432 and the cryogenic target platform 440 are coupled or include a portion of the different stages of the cryostat 400, the different vibration spectra may occur at least in part due to relative vibrations of the different stages.

(41) During operation of the cryostat 400, the distance l between the cryogenic lens assembly 412 and the target 416 is maintained constant, as described with respect to the configurations below in more details, to provide interferometric stability between the cryogenic lens assembly 412 and the target 416. The distance l between the cryogenic lens assembly 412 and the target 416 is maintained substantially constant during operation of the cryostat 400 such that a location at which the beam 420 hits the target 416 is maintained at a constant position to provide optical alignment between the cryogenic lens assembly 412 and the target 416. In order to maintain interferometric stability between the portion of the beam 420 output from (e.g., exiting) the cryogenic lens assembly 412 and the focal point on the target 416, piezoelectric actuators can be used to actuate intermediate optics, such as a dove prism (in the cryogenic lens assembly), to maintain alignment with the target 416, as shown below in FIG. 5.

(42) FIG. 5 shows a top view of the cryogenic lens assembly 412 and the target 416. As shown in FIG. 5, a right-angle prism or mirror 444 may be directly or indirectly (e.g., via support 448) fixedly coupled to the cryogenic lens assembly 412. The right-angle prism 444 is fixed relative to an optical axis A of the cryogenic lens assembly 412.

(43) A dove prism or pair of angled mirrors 452 is movably coupled to the cryogenic lens assembly 412, for example, by a support 456. An example of a dove prism that can be implemented as dove prism 452 is described in U.S. patent application Ser. No. 16/903,174, titled “Optical Alignment Using Reflective Dove Prisms,” which is incorporated herein by reference in its entirety. The right-angle prism 444 and dove prism 452 are attached to mounting structures on the cryogenic lens assembly 412 to maintain angular alignment of the beam 420 to a location on the target 416. The dove prism 452 may be movably coupled to the cryogenic lens assembly 412 by one or more piezoelectric transducers 460 that are configured to provide translation of the dove prism 452 in one or more directions to maintain the distance l . For example, the configuration illustrated in FIG. 5 includes a first piezoelectric transducer 460a configured to move the dove prism 452 in a first direction and a second piezoelectric transducer 460b configured to move the dove prism 452 in a second direction. In the present example, the piezoelectric transducers 460a, 460b may be configured to provide translation of the dove prism 452 in an X-Y plane shown by the coordinate system 464 of FIG. 5. The piezoelectric transducers 460a, 460b include a piezoelectric material. As used herein, the term “piezoelectric material” refers to a layer or material, such as a crystal or ceramic, that undergoes the converse piezoelectric effect, in which the material undergoes a displacement in response to an electric field generated by an applied electrical voltage.

Piezoelectric transducers 460a, 460b are configured to provide sub-micron displacements at high speed based on a voltage signal. Additionally, the piezoelectric transducers 460a, 460b can operate in cryogenic environments and have compact footprints. To achieve a larger operating range of the piezoelectric transducers 460a, 460b in any one direction, the piezoelectric transducers 460a, 460b can be stacked in series.

(44) As shown in FIG. 5, the beam 420 enters the cryogenic lens assembly 412 at a beam input 462 and travels through the cryogenic lens assembly 412 along the axis A of the cryogenic lens assembly 412, exiting the cryogenic lens assembly at a beam output 466. The beam 420 (e.g., the output beam) is then deflected by the prism 444 towards the dove prism 452. In the example configuration, the beam 420 is deflected by the dove prism 452 such the beam 420 is aligned with the target 416 when the beam 420 exits the dove prism 452.

(45) The piezoelectric transducers **460** are configured to be actuated to position or reposition the dove prism **452** to position or reposition the beam **420** relative to the target **416**. In particular, actuation of the piezoelectric transducers **460** compensates for relative motion between the cryogenic lens assembly **412** and the target **416**. For example, in the configuration of FIG. 5, the piezoelectric transducers **460a**, **460b** can be repositioned along the X-axis direction and Y-axis direction of the coordinate system **464**. The piezoelectric transducers **460** can compensate for fast and/or slow motion between the beam output **466** and the target **416**. An example of fast motion includes the vibrations that occur during operation of the cryostat **400**. An example of slow motion includes drift. In some aspects, a second dove prism substantially similar to the dove prism **452** may be movably coupled to the cryogenic lens assembly **412** and positioned between the beam output **466** and target **416**. In such aspects, an orientation of the second prism is substantially rotated 90 degrees about the axis A relative to the dove prism **452**. In such a configuration, the second dove prism is configured to reposition the beam **420** in a direction substantially orthogonal to the repositioning provided by the dove prism **452**.

(46) In some aspects, one or more sensors may be configured to determine the displacement between the beam output **466** and the target **416**. In some aspects, the one or more sensors may be non-contact sensors, such as, for example interferometer sensors or capacitive displacement sensors. In some aspects, other types of non-contact sensors may be used. Interferometer sensors can provide high resolution and accuracy for measuring the displacement between the cryogenic lens assembly **412** and the target **416**. Such sensors can have absolute position measurement abilities and long working distances. In configurations in which the sensor(s) are interferometric sensors, the sensor target is a reflective sensor target. In configurations in which the sensor(s) are conductive sensors, the sensor target is a conductive sensor target.

(47) There are various configurations that can be used to mount the non-contact sensors. In a first configuration, non-contact sensors can be mounted to the fixed portion of the cryogenic lens assembly **412** and aligned to a reflective or conductive sensor target (e.g., feedback reference point) near the target **416** to directly and dynamically measure x, y, and/or z displacements between the target **416** and the beam output **466**. In this configuration, one sensor per coordinate direction is used.

(48) In a second configuration, two non-contact sensors per coordinate direction are used. Both of the non-contact sensors share the same fixed reference point, such as the sensor target. In this aspect, for each coordinate direction, a first sensor is configured to determine the displacement of the cryogenic lens assembly **412** and a second sensor is configured to determine the displacement of the target **416**. Since the first and second sensors share the same fixed reference point, the relative displacement between the target **416** and cryogenic lens assembly **412** can be calculated, for example by the general controller **205** and/or the optical and trap controller **220**.

(49) FIG. 6 illustrates an example of the first configuration that is configured for determining displacements along the X-axis direction and/or Y-axis direction of the coordinate system **464**. In the configuration illustrated in FIG. 6, a first non-contact sensor **468** is coupled to the support **448** of the cryogenic lens assembly **412** and aligned with a sensor target **472** along the X-axis direction of the coordinate system **464**. The sensor target **472** is coupled to the target **416**. Interactions between the first non-contact sensor **468** and the sensor target **472** are shown schematically by the arrow B. In other aspects, the first non-contact sensor **468** may be mounted to other fixed portions of the cryogenic lens assembly **412**. A second non-contact sensor **476** is coupled to the support **448** and aligned with the sensor target **472** along the Y-axis direction of the coordinate system **464**. Interactions between the second non-contact sensor **476** and the sensor target **472** are shown schematically by the arrow C. In other aspects, the second non-contact sensor **468** may be mounted to other fixed portions of the cryogenic lens assembly **412**. Since both the non-contact sensors **468**, **476** are coupled to the cryogenic lens assembly **412** and the sensor target **472** is coupled to the target **416**, relative motion between the cryogenic lens assembly **412** and the target **416** can be

determined. In the illustrated configuration, the non-contact sensors **468**, **476** are interferometer sensors and the sensor target **472** is a reflective sensor target. In other aspects, the non-contact sensors **468**, **476** may be capacitive displacement sensors and the sensor target **472** may be a capacitive sensor target. In such aspects, the non-contact sensors **468**, **476** may be positioned closer to the sensor target **472**. For example, the capacitive sensors may be positioned within microns of the capacitive sensor target. In such aspects, the working distances of the capacitive sensors and the capacitive sensor target, and working distances of the devices, need to be selected with the estimated vibration amplitude in mind.

(50) Although the configuration of FIG. **6** shows the non-contact sensors **468**, **476** coupled to the cryogenic lens assembly **412** and the sensor target **472** coupled to the target **416**, in other configurations, the non-contact sensors **468**, **675** can be coupled to the target **416** and the sensor target **472** can be coupled to the cryogenic lens assembly **412**.

(51) FIG. **7** shows an example of the second configuration that is configured for determining displacements along the X-axis direction and/or Y-axis direction of the coordinate system **464**. The configuration shown in FIG. **7** is only described in detail herein to the extent that it differs from the configuration illustrated in FIG. **6**.

(52) As shown in FIG. **7**, a first sensor target **480** is coupled to the cryogenic lens assembly **412**. A second sensor target **484** is coupled to the target **416**. A common fixed reference point, shown schematically as block **488**, is positioned proximate the cryogenic lens assembly **412** and the target assembly **412**. A first non-contact sensor pair **492** is coupled to a portion of the fixed reference point **488** oriented such that the non-contact sensor pair **492** interacts with the first and second sensor targets **480**, **484** along the X-axis direction of the coordinate system **464**. For example, a non-contact sensor **492a** is configured to interact with the first sensor target **480** along the X-axis direction of the coordinate system **464**, as shown schematically by the arrow D. A non-contact sensor **492b** is configured to interact with the second sensor target **484** along the X-axis direction of the coordinate system **464**, as shown schematically by the arrow E. A second non-contact sensor pair **496** is coupled to a portion of the fixed reference point **488** oriented such that the non-contact sensor pair **496** interacts with the first and second sensor targets **480**, **484** along the Y-axis direction of the coordinate system **464**. For example, a non-contact sensor **496a** is configured to interact with the first sensor target **480** along the y-axis direction of the coordinate system **464**, as shown schematically by the arrow F. A non-contact sensor **496b** is configured to interact with the second sensor target **484** along the Y-axis direction of the coordinate system **464**, as shown schematically by the arrow G.

(53) Although the configuration of FIG. **6** shows the non-contact sensor pairs **492**, **496** coupled to the fixed reference point **488** and the first and second sensor targets **480**, **484** coupled to the cryogenic lens assembly **412** and the target **416**, respectively, in other configurations, the non-contact sensor pairs **492**, **496** can be coupled to the cryogenic lens assembly **412** and the target **416** and the sensor targets **480**, **484** can be coupled to the fixed reference point **488**.

(54) As described in greater detail below, active control techniques can be used to identify relative displacements between the cryogenic lens assembly **412** and the target **416** that occur based on information determined by the sensors **468**, **476**. A controller of the QIP system **200**, such as the general controller **205** and/or the optical and trap controller **220** is configured to control the piezoelectric transducers **460a**, **460b** to reposition the dove prism **452** to compensate for the relative displacement between cryogenic lens assembly **412** and the target **416**.

(55) FIG. **8** illustrates an example of beam stabilization in which the cryogenic lens assembly **412** is a fixed reference point and the target **416** shifts (e.g., is displaced) relative to the cryogenic lens assembly **412**. Such an example includes configurations in which the non-contact sensors are coupled to fixed portions of the cryogenic lens assembly **412**, such that the position of the target **416** is determined relative to the position of the cryogenic lens assembly **412**. In the example shown in FIG. **7**, shifted positions of components are indicated using the prime sign “’”.

(56) The target **416** is shown in a nominal position relative to the beam output **466**, with the beam **420** following the path shown using dash-dot lines, with a center of the target **416** shown at a position (x.sub.0, y.sub.0) at a time t_0 . During operation of the cryostat **400**, the target **416** shifts relative to the beam output **466** by x.sub.1 and y.sub.1, such that a center of the displaced target **416'** is positioned at point (x.sub.1, y.sub.1) at a time t_1 . As described in more detail below, the controller **205**, **220** may determine the displacement of the target **416'** based on position data determined by the non-contact sensors **468**, **476**. The controller **205**, **220** then commands the piezoelectric transducers **460a**, **460b** shift the dove prism **452** to account for the misalignment.

(57) FIG. **9** illustrates a flowchart of an example method **900** for dynamically repositioning a beam **420** exiting the beam output **466** of the cryogenic lens assembly **412** to compensate for relative displacements (e.g., to correct misalignments due to vibrations, drift, and so forth) between components in a QIP system **200**, such as the cryogenic lens assembly **412** and the target **416**. The method **900** is enacted by a controller of the QIP system **200**, such as, for example, the general controller **205** and/or the optical and trap controller **220** and effectively configures the system as a feedback loop configured to detect relative displacements and control the piezoelectric transducers **460a**, **460b** to reposition the dove prism **452** to maintain the path length of the beam **420** between the beam output **466** and target **416**. In some aspects, the method **900** is configured to dynamically reposition the beam **420** in real-time or substantially real time.

(58) Initially, at **904**, the cryogenic lens assembly **412** is aligned with the target **416** such that the output beam is aligned with the target **416**.

(59) At **908**, the controller **205**, **220** determines the initial position of the target **416** based on position information determined by the non-contact sensors **468**, **476**. In some aspects, the first position of the target **416** is determined based on a reference point, such as a portion the cryogenic lens assembly **412**.

(60) At **912**, the controller **205**, **220** determines a second position of the target **416** based on position information determined by the non-contact sensors **468**, **476**. In some aspects, the controller **205**, **220** may be configured to determine the second position of the target **416** after a predefined amount of time has elapsed since the first position of the target **416** was determined. In some aspects, the controller **205**, **220** may monitor the position of the target **416** continuously, and the term “second position” refers to any position of the target **416** that is determined after a previous position of the target **416** has been determined.

(61) At **916**, the controller **205**, **220** determines a displacement of the target **416** between the first position and the second position.

(62) At **920**, the controller **205**, **220** determines an amount to shift the dove prism **452** to maintain alignment between the beam **420** exiting the beam output **466** and the target **416** to realign the beam **420** with the target **416**. For example, the controller **205**, **220** may determine a the amount to shift the dove prism **452** based on the displacement of the target **416** determined at **816** to align the beam **420** with the target.

(63) At **924**, the controller **205**, **220** generates a repositioning signal based on the determined amount to shift the dove prism **452**.

(64) At **928**, the controller **205**, **220** shifts the dove prism **452** in the X-axis direction and Y-axis direction by half the distance that the target **416'** has moved in each coordinate direction relative to the beam output **466** to reposition the beam **420**.

(65) As described above, piezoelectric transducers are actuators that can be configured as the drive mechanism to correct for vibration disturbances. Piezoelectric actuators provide sub-micron displacements at high speed; limitations are imposed by the frequency and noise capabilities of the voltage signals as opposed to the mechanism of the device. An alternative drive mechanism for stabilizing the beam output of a cryogenic lens is to mount it on bearing-based nanopositioners. However, not only must the bearing-based nanopositioners support heavy loads of the cryogenic lens assemblies, which can be on the order of 500 g, but the stability and rigidity of the assembly

will be significantly less than what is provided by the piezoelectric actuators, which are solid ceramic bodies. In the application of the piezoelectric actuators, the stiffness limitations will be at the boundaries between the lens, mounting structures, piezos (i.e., piezoelectric actuators/transducers), and dove prism as opposed to the piezos themselves. Additionally, the piezoelectric actuators can operate in cryogenic environments and have compact footprints. To achieve a larger operating range of the piezoelectric actuator in any one direction, multiple devices can be stacked in series, making the solution versatile for many operating conditions.

(66) The previous description of the disclosure is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the common principles defined herein may be applied to other variations without departing from the scope of the disclosure. Furthermore, although elements of the described aspects may be described or claimed in the singular, the plural is contemplated unless limitation to the singular is explicitly stated. Additionally, all or a portion of any aspect may be utilized with all or a portion of any other aspect, unless stated otherwise. Thus, the disclosure is not to be limited to the examples and designs described herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

Claims

1. A system for piezoelectric-based beam stabilization, the system including: a laser beam source configured to produce a laser beam; a target; a lens assembly including a lens objective configured to produce an output beam from the laser beam and at least one optical element movably coupled to the lens assembly and configured to direct the laser beam to the target; and one or more piezoelectric transducers coupled to the at least one optical element and configured to dynamically reposition the at least one optical element in response to relative motion between the target and the lens assembly.
2. The system of claim 1, wherein the at least one optical element includes a dove prism.
3. The system of claim 1, wherein the at least one optical element includes a first optical element, and further comprising a second optical element fixedly coupled to the lens assembly, the second optical element configured to direct the output beam onto the first optical element.
4. The system of claim 1, further comprising one or more non-contact sensors coupled to one of the target and the lens assembly and a sensor target coupled to the other of the target and the lens assembly.
5. The system of claim 4, wherein the one or more non-contact sensors include at least one of interferometer sensors, capacitive sensors, or a combination thereof.
6. The system of claim 4, further comprising a controller configured to: receive information indicative of a position of the position of the target or the lens assembly from the one or more non-contact sensors; determine a displacement of one of the target and the lens assembly relative to the other of the target and the lens assembly; and command the one or more piezoelectric transducers to reposition the at least one optical element to reposition the laser beam relative to the target.
7. The system of claim 1, wherein the target and the lens assembly are fixedly coupled to a common surface, and wherein the relative motion is caused by vibrations of the common surface or thermal contractions in one or more of the target and the lens assembly.
8. The system of claim 1, wherein the lens assembly is fixedly coupled to a first surface and the target is fixedly coupled to a second surface, and wherein the relative motion is caused by relative vibrations between the first surface and the second surface.
9. The system of claim 8, wherein the first surface and the second surface are positioned inside a cryogenic environment.
10. The system of claim 1, wherein the system is a quantum information processing (QIP) system including a cryostat configured to form a cryogenic environment.

11. The system of claim 10, wherein the target includes an ion trap.
 12. The system of claim 10, wherein the laser beam source is positioned outside of the cryogenic environment.
 13. A computer implemented method for performing piezoelectric-based beam stabilization for a cryogenic environment, the computer-implemented method comprising: receiving, by a computing system including a processor and a memory, information indicative of a relative position of a target and a cryogenic lens assembly positioned inside the cryogenic environment; determining, by the computing system, an amount of displacement between the target and the cryogenic lens assembly; and generating, by the computing system, a repositioning command configured to actuate one or more piezoelectric elements coupled to a movable optical element of the cryogenic lens assembly to reposition the movable optical element to direct an output beam produced by an objective lens of the cryogenic lens assembly onto the target.
 14. The method of claim 13, wherein performing the piezoelectric-based beam stabilization for the cryogenic environment includes performing as part of a quantum information processing (QIP) system.
 15. The method of claim 13, wherein the information indicative of the relative position of the target and the cryogenic lens assembly is determined by one or more non-contact sensors coupled to one of the target and the cryogenic lens assembly and a sensor target coupled to the other of the target and the lens assembly.
 16. The method of claim 15, wherein the one or more non-contact sensors include at least one of interferometer sensors, capacitive sensors, or a combination thereof.
 17. The method of claim 13, wherein the target includes an ion trap.
 18. The method of claim 13, wherein the method occurs in real-time or in substantially real-time.
 19. The method of claim 13, wherein the target and the cryogenic lens assembly are fixedly coupled to a common surface, and wherein relative motion between the target is caused by vibrations of the cryostat or thermal contractions in one or more of the target and the lens assembly.
 20. The method of claim 13, wherein the cryogenic lens assembly is fixedly coupled to a first surface and the target is fixedly coupled to a second surface, and wherein relative motion is caused by relative vibrations between the first surface and the second surface.
-